

**LOCAL BUCKLING AND CRIPPLING OF
BEAMS/STRINGERS**

Part: **3.4 – STRENGTH OF COMPOSITE PARTS**

Chapter: **3430**

Title: **LOCAL BUCKLING AND CRIPPLING OF BEAMS/STRINGERS**

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1 INTRODUCTION**1.1 OBJECT**

This chapter is concerned with the local buckling stress of composite thin walled sections when loaded in compression, as well as with the ultimate load the section (in a short length) is capable to sustain before failure beyond local buckling, i.e. with the crippling strength.

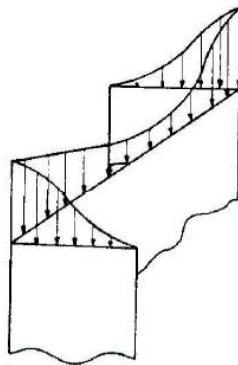


Figure 1. *Typical compression stress distribution in a thin-walled profile after local buckling*

For sustaining compression loads materials are formed and built in order to get composite shapes such as angles, channels, zees, etc. with significantly higher buckling stresses than simple flat sheets. Calculation methods for predicting both the local buckling stress and the ultimate compression strength (crippling) of the section normally decompose the section into flat elements, supported on one or two sides, before using available theories for evaluating buckling and crippling stresses in flat plates.

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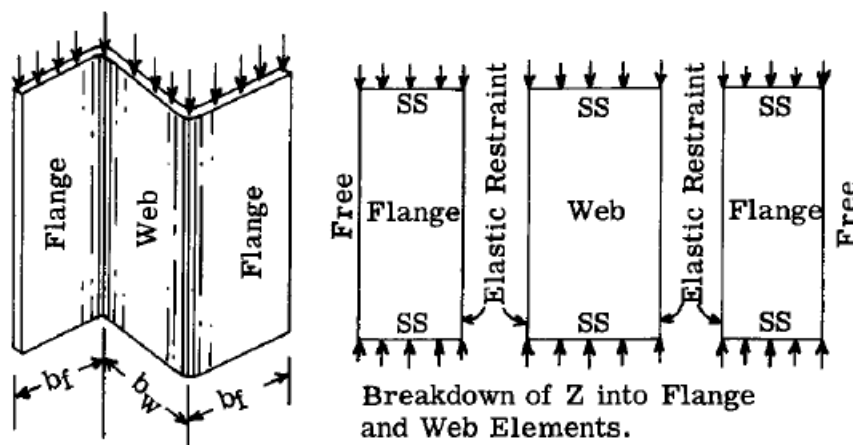


Figure 2. Typical decomposition of a thin walled section into flat elements (Bruhn C6.3)

A section element with one of its sides supported and the other free is typically referred as a flange (or flange element). In Z, J or I shaped sections, the centre element supported on both sides is typically referred as the web (or web element).

The crippling strength of metallic profiles was treated in chapter 3130 of this Manual. Indeed many paragraphs and explanations are common for both types of materials and have been literally copied to this chapter.

1.2 SCOPE AND LIMITATIONS

The method has the following limitations:

- ✓ Composite profiles with thin walled sections
- ✓ Short column range (column buckling not covered herein, see chapter 3440)

2 LOCAL BUCKLING IN COMPOSITE BEAMS

As shall be discussed in following chapter, compression members are capable to sustain load beyond the local buckling onset (of one or more section elements). However, the calculation of the local buckling stress of a section may be of interest for several reasons, including:

- ✓ Variable compression stress across element width (crippling methods only deal with uniform stresses and may be incapable to estimate allowable loads with shear/bending interaction)
- ✓ Setting a cut-off value for allowable stresses in certain types of sections acting as column members with risk of global buckling (e.g. L or T stringers of a stiffened panels)
- ✓ Limiting the initial onset of buckling and resulting load distribution to certain load levels according to a buckling policy
- ✓ Etc...

Several references include solutions, either exact or approximated, for calculating the buckling loads of the segments of the composite sections. Among others Lekhnitskii, HSB and the CMH-17 are references worth to cite. The support condition on supported edges shall be simple supports (clamping

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edges or introducing rotation restraints in calculations shall not be permitted without further explicit validations).

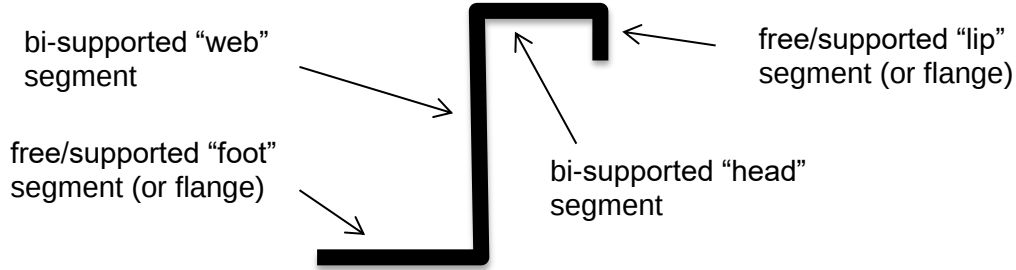


Figure 3. Typical decomposition of a thin walled cross section in segments

In following pages, the beam/column axis shall be considered aligned with X axis and this used as reference for the orientation of the composite material plies in the laminates (i.e. X aligned, as usual, with the warp direction of the plies oriented at 0°).

All solutions in literature are presented for orthotropic laminates, those without bending/torsion coupling in the $[D]$ stiffness matrix, i.e. with $D_{16} = D_{26} = 0$. For typical symmetrical and balanced laminates, this will be strictly true if made of bi-axial fabrics (materials with $E_1=E_2$) and a reasonably good approximation if made of unidirectional tapes with regularly distributed layer orientations, except for really thin laminates.

$$[D] = \begin{bmatrix} D_{11} & D_{12} & 0 \\ D_{12} & D_{22} & 0 \\ 0 & 0 & D_{66} \end{bmatrix}$$

For laminates slightly unbalanced or unsymmetrical, the terms of “generalized” bending stiffness matrix $[D^*]$ shall be used instead in following expressions, where:

$$[D^*] = [D] - [B] \cdot [A]^{-1} \cdot [B]$$

Additionally, the detrimental effect of the through-thickness (or transverse) shear stiffness in buckling stability is neglected in these formulations. The shear stiffness is assumed infinite and the stability driven by bending, what it is a good approximation for thin walled sections ($\sim b/t > 25$).

2.1 BUCKLING IN COMPRESSION

Bi-supported segment in compression

From HSB 45111-08 or from CMH-17 Vol. 3, chapter 9, equation 9.2.1.3 (assumed column length a is high compared with flange width b ; $\sim a/b > 3$):

$$N_{xb} = (2 \cdot \pi^2 / b^2) \cdot [(D_{11} \cdot D_{22})^{0.5} + D_{12} + 2 \cdot D_{66}]$$

... is the buckling compression load-flow (force per unit width) of a bi-supported segment of a cross section profile made of fibrous composite laminates.

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Compression buckling of non-orthotropic laminates in bi-supported long plates

In thin laminates made of UD tape materials, even those following the typical design rules, bending-torsion coupling effects may be significant and should not be neglected. Herencia, Weaver and Friswell derived expressions for calculating the compression buckling load of long plates with flexural/twist anisotropy (Ref. 11):

$$N_{xb} = K_x \cdot \pi^2 \cdot (D_{11} \cdot D_{22})^{1/2} / b^2$$

Where the buckling coefficient K_x , for simply supported conditions, is:

$$K_x = 2 \cdot (1 - 4 \cdot \delta \cdot \gamma - 3 \cdot \delta^4 + 2 \cdot \delta^2 \cdot \beta)^{1/2} + 2 \cdot (\beta - 3 \cdot \delta^2)$$

...expressed in terms of the non-dimensional parameters suggested by Nemeth:

$$\begin{aligned} \alpha &= (D_{22}/D_{11})^{1/4} & \beta &= (D_{12} + 2 \cdot D_{66}) / (D_{11} \cdot D_{22})^{1/2} \\ \gamma &= D_{16} / (D_{11}^3 \cdot D_{22})^{1/4} & \delta &= D_{26} / (D_{11} \cdot D_{22}^3)^{1/4} \end{aligned}$$

Flange in compression with one edge free

From HSB 45111-08 or from CMH-17 Vol. 3, chapter 9, equation 9.2.1.5 (assumed column length a is high compared with flange width $b \sim a/b > 3$):

$$N_{xb} = 12 \cdot D_{66} / b^2 + \pi^2 \cdot D_{11} / a^2 \quad (\text{one lateral side free, rest simply supported})$$

... is the buckling compression load-flow (force per unit width) of a supported/free segment (flange) of a cross section profile made of a fibrous composite orthotropic laminate (D_{16} , D_{26} terms negligible).

In case the flange ends have more effective support conditions, being still the long sides supported/free, the beam length in the second term can be replaced by the effective column length (see Ref. 6). In case of clamped ends:

$$N_{xb} = 12 \cdot D_{66} / b^2 + 4 \cdot \pi^2 \cdot D_{11} / a^2 \quad (\text{one lateral side free, the other simply supported, ends of the plate clamped})$$

In case the column length is sufficiently high ($a/b \gg 1$), the second term becomes negligible, thus:

$$N_{xb} = 12 \cdot D_{66} / b^2$$

Compression buckling of anisotropic long flanges with one edge free and the other simply supported

Weaver and Herencia obtained solutions for the compression buckling load of long plates made of laminates with flexural anisotropy, one side free and the other simply supported (Ref. 10). They suggested to take, as critical compression buckling load, the minimum of the following expressions:

$$\begin{aligned} N_{xb1} &= 12 \cdot (D_{66} - D_{26}^2/D_{22}) / b^2 \\ N_{xb2} &= (12 \cdot \varepsilon - 10 \cdot \gamma^2) \cdot (D_{11} \cdot D_{22})^{1/2} / b^2 \end{aligned}$$

Where ε and γ are the following non-dimensional parameters:

$$\begin{aligned} \varepsilon &= D_{66} / (D_{11} \cdot D_{22})^{1/2} \\ \gamma &= D_{16} / (D_{11}^3 \cdot D_{22})^{1/4} \end{aligned}$$

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In other words, the ratio between the compression buckling of the long flange, with one side simply-supported and the other free, relative to that neglecting flexural anisotropy would be, then:

$$F_{ani} = \min(N_{xb1}, N_{xb2}) / (12 \cdot D_{66} / b^2)$$

This factor will be used to correct following solutions for compression buckling, only available for orthotropic laminates.

Compression buckling of orthotropic long flanges with one edge free and the other clamped

The HSB45111-10 (§ 3.2) proposes an approximation for this case as a closed form expression:

$$N_{xb} = K_x \cdot \pi^2 \cdot (D_{11} \cdot D_{22})^{1/2} / b^2$$

Where the buckling coefficient K_x , for clamped-free conditions, is:

$$K_x = 0.72 + 0.88 \cdot \beta - 0.99 \cdot \beta \cdot \zeta - 0.03 \cdot \beta^2$$

...and β and ζ are the following non-dimensional parameters:

$$\beta = (D_{12} + 2 \cdot D_{66}) / (D_{11} \cdot D_{22})^{1/2}$$

$$\zeta = D_{12} / (D_{12} + 2 \cdot D_{66})$$

Compression buckling of orthotropic long flanges with one edge free and the other elastically restrained (by other segment of a beam)

The same HSB45111-10 (sections § 3.1 and § 3.3) proposes solutions also for intermediate support conditions on the non-free edges: for the flange of an I-section profile and for the flange of a C or Z profile.

It could be practical to extend the capability of this methodology for a T stringer co-cured or bonded to a skin panel. In such case, we'd use the solution for the flange of the C/Z profile (the skin between stringers shall correspond to the "web" in those expressions), but doubling the value of the clamping stiffness $c_{\phi,C}$ in expression (9) of Ref. 12 (one web/skin-panel at each side of the stringer T-flange).

Compression buckling of orthotropic long plates with one edge free and other rotationally restrained

Tarján, Sapkás, Kollár (Ref. 14) collected approximated solutions in closed form for buckling of long plates relevant for the calculation of the local buckling of a composite beam. In particular, for segments with one edge free and the other rotationally restrained, the approximation suggested, assuming the support element is stable, would be:

$$N_{xb} = [7 / (1 + 4.12 \cdot \zeta)] \cdot [(D_{11} \cdot D_{22})^{0.5} / b^2] + 12 \cdot D_{66} / b^2$$

$$\zeta = D_{22} \cdot b / GJ_s$$

... where GJ_s is the torsion stiffness of the support

$$GJ_s = 4 \cdot [b' \cdot (D'_{66} - D'^2_{26} / D'_{66})]_s$$

... in case the support is a strip of a laminate of width b' and bending stiffness matrix $[D]'$

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...where the s subscript refers to the area providing the rotational restraint to the supported side of the flange. For the web of a T stringer cocured/bonded to a skin panel, its compression buckling can be calculated this way, considering the support area is the sum of the skin-pad and stringer-foot laminates (see figure).

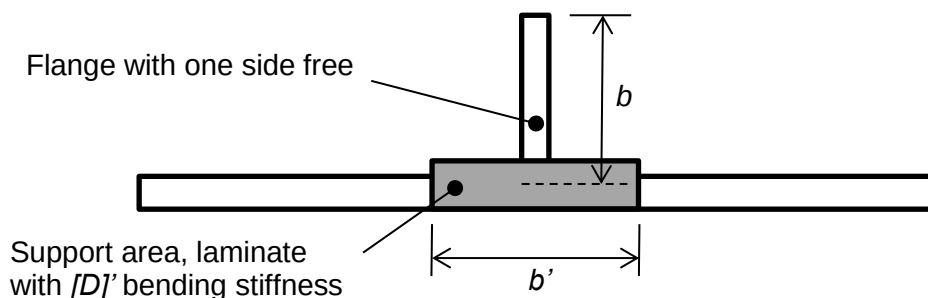


Figure 4. Flange/web of T stringer cocured/bonded to a skin panel

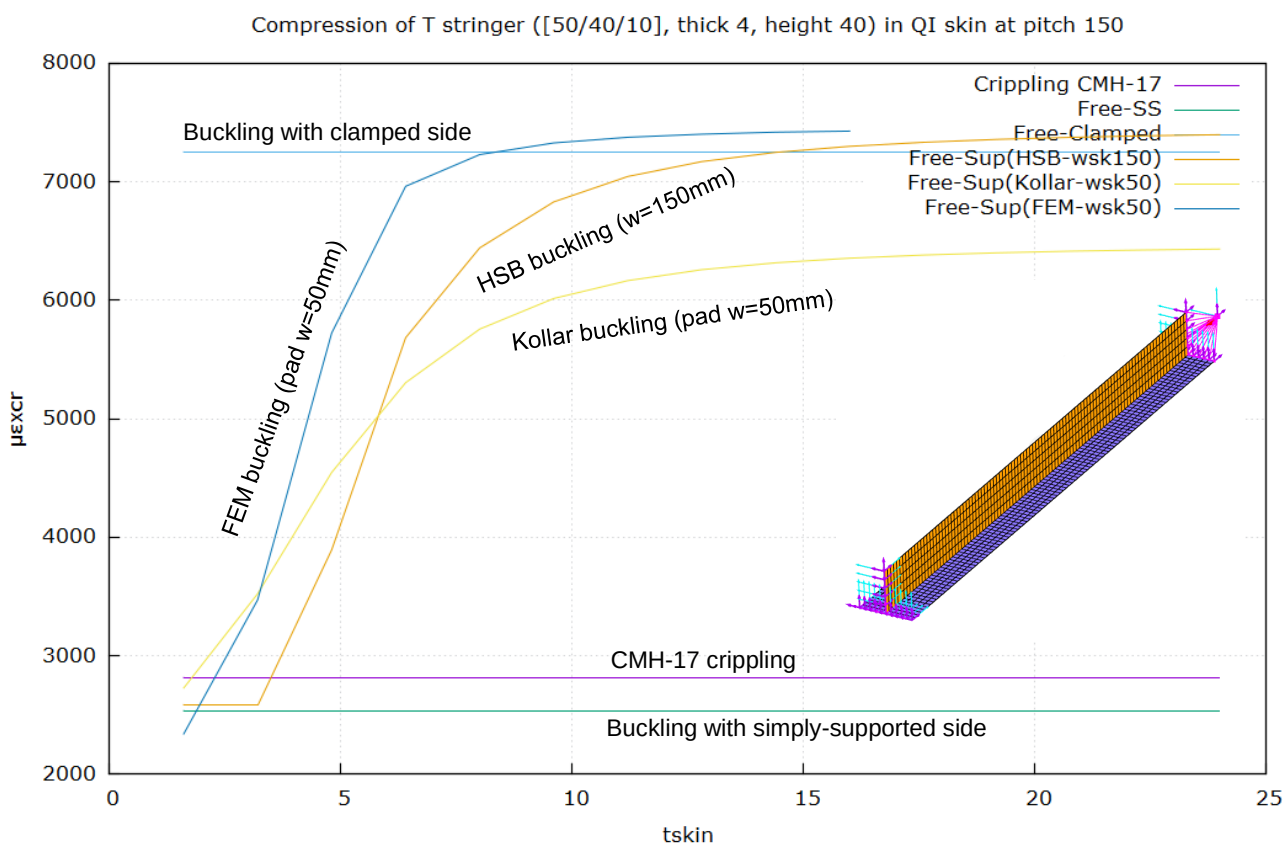


Figure 5. Predictions for compression buckling and crippling in the flange of a T stringer

For OEF flanges in C, Z or J section profiles (supported in “web” segments) the HSB formulation is recommended. For the web of T section profiles stiffening skin panels, the formulation from Kollar is preferred, using the rotational restraint of the skin pad plus stringer foot laminate within a limited width.

For bi-supported segments no closed form solutions different than the one for simple-supports condition are presented herein.

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2.2 BUCKLING IN SHEAR

Bi-supported segment in shear

A solution for buckling under pure shear can be found in literature. A graphical handbook solution based on the results of Seydel and Davenport and Bert is presented in the HSB 45112-02. Note that α is the effective aspect ratio of the plate and β is Seydel's orthotropy parameter:

$$\alpha = b/a \cdot (D_{11} / D_{22})^{1/4}$$

$$\beta = (D_{12} + 2 \cdot D_{66}) / (D_{11} \cdot D_{22})^{1/2}$$

With the buckling factor k obtained from the HSB 45112-02 figure, the critical buckling in-plane section shear force q_{cr} per unit width is calculated as:

$$q_{cr} = k \cdot (\pi / b)^2 \cdot (D_{11} \cdot D_{22}^3)^{1/4}$$

The in-plane mean shear stress can be obtained with: $\tau_{cr} = q_{cr}/t$

For the frequent case of high aspect ratios, the formula of a plate with $\alpha = 0$ (infinite plate) is given below (small conservative error for $a/b \geq 3$):

$$\beta > 1: \tau_{cr} = 4 / (t \cdot b^2) \cdot (D_{22} \cdot (D_{12} + 2 \cdot D_{66}))^{1/2} \cdot (11.7 + 0.532 / \beta + 0.938 / \beta^2)$$

$$\beta \leq 1: \tau_{cr} = 4 / (t \cdot b^2) \cdot (D_{11} \cdot D_{22}^3)^{1/4} \cdot (8.12 + 5.05 \cdot \beta)$$

Shear buckling of non-orthotropic laminates in bi-supported long plates

Again, in thin laminates made of UD tape materials, bending-torsion coupling effects may be significant and should not be neglected.

Nemeth (Ref. 9) proposed solutions for this problem in long flat laterally supported plates. In the case of shear buckling, it is possible to apply a correction factor to the buckling load obtained as specially orthotropic plate, interpolating in graphs with the parameters defining the flexural-twisting anisotropy in the laminate stiffness matrix $[D]$ (see following figure).

It is recommended to use conservatively the correction factors proposed for positive shear in the graph, unless the critical values of positive and negative shear are significantly different (and thus the shear sign a design driver) and the conventions for signs and laminates stacking are fully consistent and controlled (in design and production).

A closed form solution for this problem was proposed later by Weaver (Ref. 13):

$$N_{xyb} = K_{xy} \cdot \pi^2 \cdot (D_{11} \cdot D_{22}^3)^{1/4} / b$$

$$\text{with: } K_{xy} = 3.42 + 2.05 \cdot \beta - 0.13 \cdot \beta^2 - 1.79 \cdot \gamma - 6.89 \cdot \delta + 0.36 \cdot \beta \cdot (2 \cdot \gamma + \delta) - 0.25 \cdot (2 \cdot \gamma + \delta)^2$$

Note the sign of γ and δ changes if the angle of the plies ($\pm 45^\circ$) changes, or if the sign of the shear load changes. Again, it is recommended that the minimum value obtained with $+\gamma$ and $+\delta$ and with $-\gamma$ and $-\delta$ shall be used in order to be conservative.

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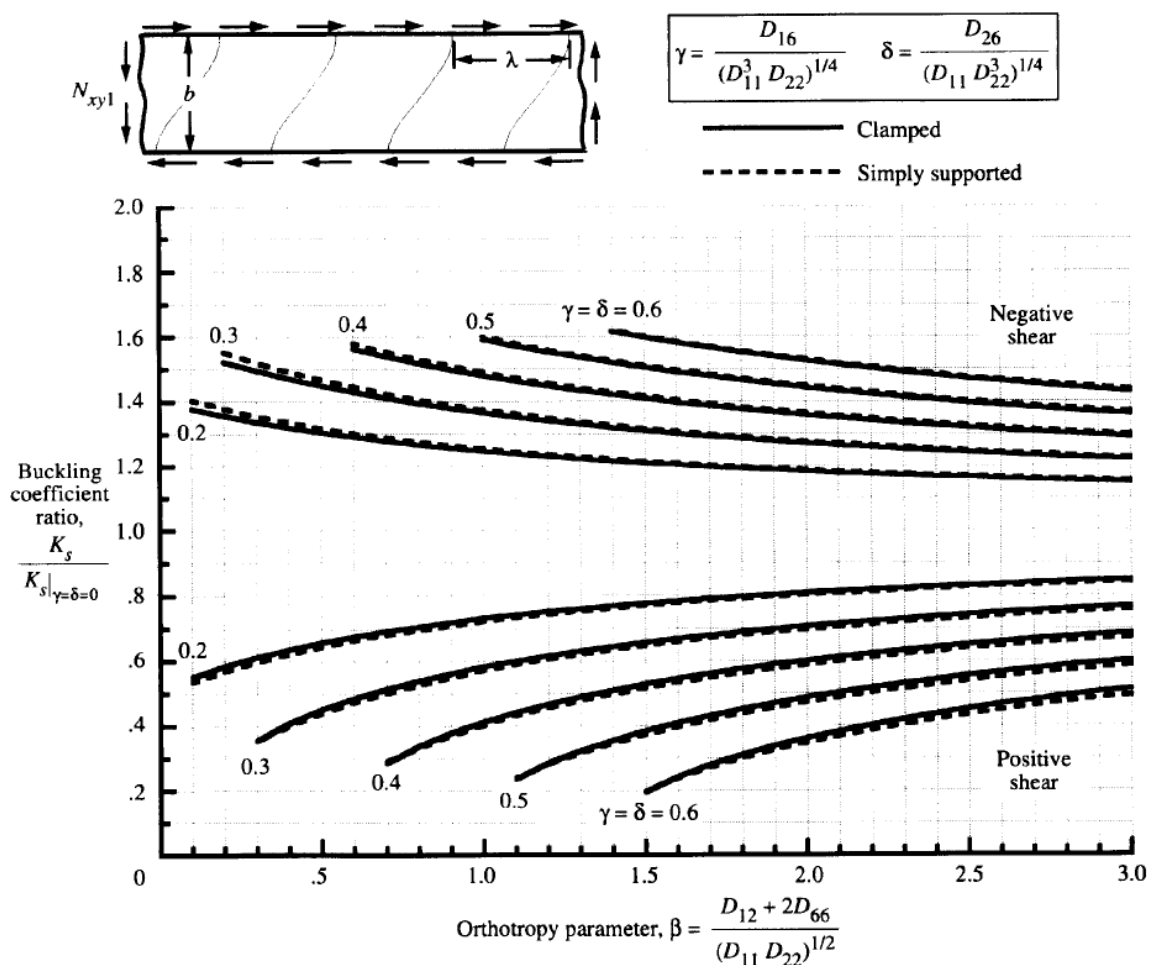


Figure 6. Effect of anisotropy on buckling coefficients for long flat plates in shear loads (Ref. 9)

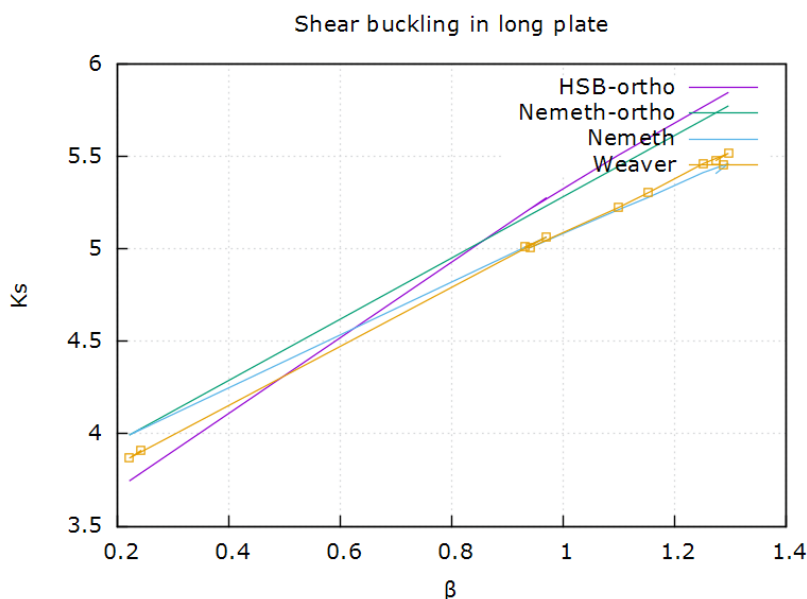


Figure 7. Solutions for shear buckling K_s factors for laminates made of UD of increasing β

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2.3 BUCKLING IN IN-PLANE BENDING

For the case of buckling under linearly distributed loading, a graphical handbook solution is presented in HSB 45111-09. The parameters α and β are defined in the same way as above. The results of HSB 45111-09 have been created using Ritz method with a high number of trial functions.

Lekhnitskii proposes a closed form analytical solution which was also obtained by Ritz method using only two trial functions and thus leading to non-conservative results (~5%). However, this analytical approximation might be useful for quick initial design checks and is given below for the special case of a pure in-plane bending. One can directly use the following equations from Lekhnitskii's book by iterating over the half wave m in order to obtain the critical buckling moment:

$$M_{cr} = \frac{\pi^2 \sqrt{D_{11} D_{22}}}{6} k$$

With:

$$k = \frac{9\pi^2}{32} \sqrt{a_{m1} a_{m2}}$$

$$a_{m1} = \sqrt{\frac{D_{11}}{D_{22}}} \left(\frac{m}{c}\right)^2 + \frac{2(D_{12} + 2D_{66})}{\sqrt{D_{11} D_{22}}} + \sqrt{\frac{D_{22}}{D_{11}}} \left(\frac{c}{m}\right)^2$$

$$a_{m2} = \sqrt{\frac{D_{11}}{D_{22}}} \left(\frac{m}{c}\right)^2 + \frac{8(D_{12} + 2D_{66})}{\sqrt{D_{11} D_{22}}} + 16 \sqrt{\frac{D_{22}}{D_{11}}} \left(\frac{c}{m}\right)^2$$

$$c = \frac{a}{b}$$

The minimum value of k can be found for $c = \frac{m}{\sqrt{2}} \sqrt[4]{D_{11}/D_{22}}$ and thus for the infinite plate $c \rightarrow \infty$ with:

$$k = 11.1 \cdot \left(1.25 + \frac{D_{12} + 2D_{66}}{\sqrt{D_{11} D_{22}}} \right)$$

For the final design, it is mandatory to use the values given by HSB 45111-09.

2.4 BUCKLING UNDER COMBINED LOADS

Abovementioned reference records typical interaction formulas used to obtain the safety margin vs buckling stability in multi-axial stress states. The proposed failure law is:

$$R_c + R_f^{1.76} + R_s^2 = 1$$

...where R_c , R_f and R_s are the failure indices in uni-axial loads for axial longitudinal compression, in-plane bending (flexure) and shear respectively.

That expression can be expressed introducing the Reserve Factor for performing the strength check of a general load case:

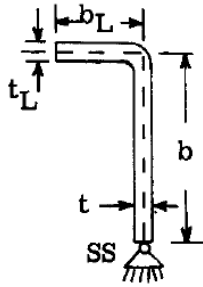
$$(RF \cdot N_x / N_{xb}) + (RF \cdot M / M_b)^{1.76} + (RF \cdot N_{xy} / N_{xyb})^2 = 1$$

If the in-plane applied bending M is not zero, the Reserve Factor shall have to be solved numerically.

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2.5 EFFECTIVE LATERAL SUPPORT

Section elements will be effectively supported on one of its sides by other element/flange only if this supporting (perpendicular) flange is stiff enough. The check may be done using *Windenburg's* methodology (see Bruhn C7.9), after an appropriate adaptation to composite laminates. If t is the thickness of the (first) element, b its equivalent width (using midline crossings as end points), A_L and I_L the area and moment of inertia of the lateral flange intending to provide support, this support shall be effective if:



Area and inertia of flange intending to provide effective support:

$$A_L = b_L \cdot t_L$$

$$I_L = (t_L \cdot b_L^3) / 12 + A_L \cdot (b_L/2)^2 = (t_L \cdot b_L^3) / 3$$

$$3(1-\nu_{12}^2) \cdot I_L / (b \cdot t^3) - A_L / (b \cdot t) \geq 5$$

Figure 8. Parameters in Windenburg's check for effective lateral support

This original formula is only valid for a stiffener made from the same homogenous material where the dependency of the E-Modulus disappears. By using the equivalent E-Moduli along the longitudinal axis of the stiffener (x-axis), the Windenburg formula can be re-written for composite materials as:

$$(E_{x,L} \cdot t_L \cdot b_L^3 / 3) / (D_{11} \cdot b)_{web} - 4 \cdot (E_{x,L} \cdot t_L \cdot b_L) / (E_x \cdot b \cdot t)_{web} \geq 20$$

In previous expression E_L , t_L , b_L are the apparent Young modulus (at the profile axis direction), thickness and width for the lip intending to provide support, E , t , b idem for the segment that has to be laterally supported, and finally D_{11} the bending term of supported plate.

For generic section shapes not actually covered by dedicated formulations accounting implicitly for effective support (e.g. C, Z or J section profiles in HSB sheets, see discussion on previous pages), this check shall be done.

3 CRIPPLING OF COMPOSITE BEAMS

3.1 INTRODUCTION

Postbuckling is the ability of a compressive member or stiffened panel to carry loads well in excess of the initial buckling load. The "postbuckling range" may be considered to exist between the initial buckling load and some higher load representing failure, e.g., delamination at the free edge of a compressive member or the disbonding of a stiffener from the panel in a stiffened panel. ... For composite panels, tangent stiffnesses are a function of material properties and lay-up. Local buckling of one or more of the plate elements comprising a stiffener will similarly reduce the in-plane stiffnesses of the affected elements and will cause the load to shift to the unbuckled portions of the stiffener. The upper limit of the postbuckling range is sometimes referred to as "local crippling" or simply "crippling".

Compressive crippling is a failure mechanism in which the cross section of a stiffener is first loaded in compression and becomes distorted in its own plane without translation or rotation of the entire column taking place. Then, ultimate failure (crippling) results from block compression strength or compressive

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instability of the unbuckled portions of the column cross-section, and may be preceded by delaminations induced by the post-buckled deformations.

Previous paragraphs are extracts from CMH-17 Volume 3 chapter 9.2.2. Practical methods for predicting the crippling strength of composite sections are empirical or semi-empirical. Also as done in metallic sections, calculation of the crippling strength is preceded by a decomposition of the section in flat segments, being the total crippling strength the sum of the crippling loads of those segments.

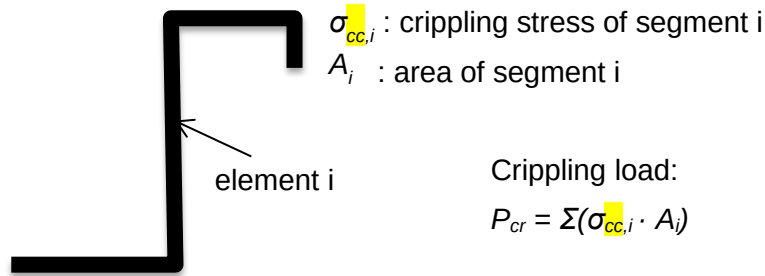


Figure 9. Crippling load as sum of the crippling loads of the section elements/flanges

3.2 CRIPPLING OF BEAM SECTION ELEMENTS/SEGMENTS

That same reference proposes approximations for the failure law of composite sections segments loaded in compression in terms of their crippling (apparent) stress F_{cc} , as function of the laminate plain compression strength F_{cu} , and the width to thickness ratio (b/t).

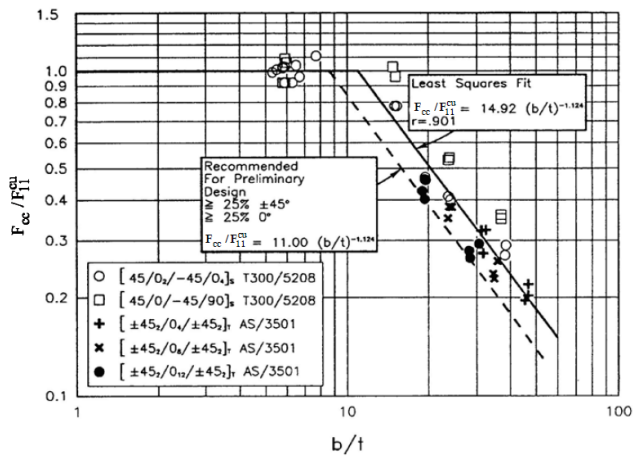


FIGURE 9.2.2(f) Normalized crippling data - no-edge-free.

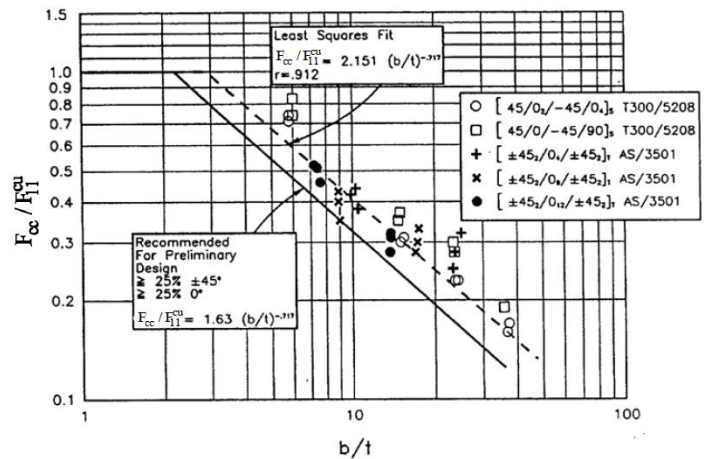


FIGURE 9.2.2(h) Normalized crippling data - one-edge free.

Figure 10. Crippling of composite section segments per CMH-17 Vol. 3

Those crippling predictions, at least the recommended for preliminary design, can be overly conservative for typical composite beam sections, however (see Ref. 15). Spier (Ref. 6) proposed a potential law as approximation model for the crippling strength of the segment F_{cc} , relating this value with the local buckling load of the segment F_{cb} and the laminate plain strength in compression F_{cu} .

$$(F_{cc} / F_{cb}) = \alpha \cdot (F_{cu} / F_{cb})^n$$

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This approximation has been used as legacy method within Airbus in the past (see Ref. 7). It offers a significant advantage: referencing the crippling strength with the buckling load provides consistency in predictions of those two properties (buckling and crippling). However, the legacy values of the adjustment parameters of the model (α and n) were not validated except for a few composite materials. A later study of available tests data (legacy data plus recent tests conducted within Airbus DS, and data reported in Spier's paper) has been performed and concluded in a set of design values valid for stress sizing and justification of generic composite beams (see details in Ref. 15) which, conveniently presented in terms of allowable strains, becomes:

$$\varepsilon_{cc} / \varepsilon_{cu} = \min [1.0 , \alpha \cdot (\varepsilon_{cb} / \varepsilon_{cu})^{1-n}]$$

Where:

ε_{cc} Crippling strain of the segment

ε_{cu} Plain compression allowable strain of the material

ε_{cb} Buckling compression strain of the segment

$\alpha = \begin{cases} 0.50 & \text{if segment is supported at its both sides (NEF)} \\ 0.63 & \text{if segment is supported at just one of its sides (1EF, i.e. it is a flange segment)} \end{cases}$

$n = 0.61$

Effective support

The method that shall be used (either if the adjacent element intending to provide support is another flange or a bi-supported element) shall be the *Windenburg's* check presented in previous sub-chapter (see §2, and Figure 8).

If this check is not true, the support condition of that lateral side shall be set to free, and crippling stress of the element calculated accordingly.

If both sides of a section element become free after the checks of effective lateral support, the crippling stress of the element shall be set to 0 (as much the column buckling strength of the element without lateral supports).

Lips

Reinforcement lips at the edge of "free" lateral sides should be treated as flanges. Effective support of main element onto the lip/flange element (and vice-versa) shall be checked before calculating final crippling strengths. The support condition of the main element will be changed to bi-supported if lip provides effective support. If lip is not effective, the lip area shall be accounted to determine the final crippling strength, and may even change the support condition of the side to supported

Statistical basis of stiffness in buckling/crippling calculations

B-basis (compression) values of the material elastic moduli shall be used to compute the bending stiffness of the laminates in buckling calculations. Material strengths shall be also considered as B-basis as usual (*).

However, for normalizing buckling/crippling loads/stresses into strains, the mean/nominal values of the elastic moduli shall be used instead.

(*) Nevertheless, only for calculating the ratio ($\varepsilon_{cb} / \varepsilon_{cu}$) in crippling (Spier's) formula, mean-values for material stiffness and strength is permitted.

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3.3 CRIPPLING FAILURE LOAD OF THE COMPOSITE SECTION

3.3.1 General case

As discussed previously, the allowable crippling load of the composite beam section can be calculated as the sum of the crippling loads of its segments.

3.3.2 Crippling of cocured/bonded Skin Stringers

Some special considerations are required to predict correctly the compression strength of a composite stringer which is joined to a skin panel, either by rivets or cocured or bonded. For calculating their compression strength, the following conventions and policies shall apply (taken from company's legacy methods, Ref. 7 and Ref. 8).

Stringer foot riveted to skin

If stringer is riveted to skin it shall be considered conservatively that the stringer is completely independent, and crippling of stringer foot segments shall be calculated taking into account exclusively the stringer laminates.

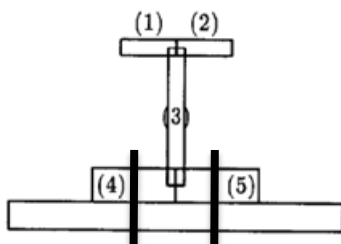


Figure 11. Stringer riveted to skin (crippling calculated independently of skin)

Stringer foot cocured/bonded with skin

- ✓ If the total thickness of the skin pad plus the bonded stringer foot is lower than twice the thickness of the adjacent skin panels, then the skin+foot segments shall NOT be included in the crippling calculation, but considered part of the skin.

$$t_{pad} + t_{str} < 2 \cdot \max(t_1, t_2)$$

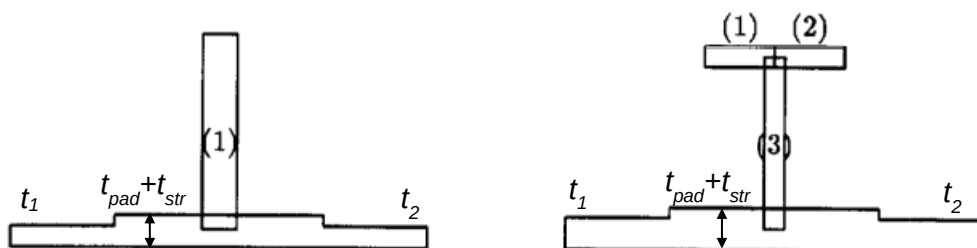


Figure 12. Stringer bonded to skin with foot thinner than skin

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- ✓ If the total thickness of the skin pad plus the bonded stringer foot is higher than twice the thickness of the adjacent skin panels, then the skin+foot segments shall be included in the crippling calculation (as supported/free segments).

$$t_{pad} + t_{str} \geq 2 \cdot \max(t_1, t_2)$$



Figure 13. Stringer bonded to skin with foot thicker than skin

Tee stringers

The stringer crippling strength would be typically used as cut-off value for calculating the global buckling load of the stiffened panel composed by the skin and its stringers. However, it cannot be expected that the post-buckled web segment of a tee (or L-shaped) stringer provides bending inertia to the stiffened panel, thus...

The cut-off value for the compression strength of a tee stringer stiffening a panel shall be the minimum between the local buckling local and the crippling strength of the stringer web segment.



Figure 14. T-stringer stiffened panel

The proposed approach sets the minimum between buckling and crippling, and not just the buckling load, in order to stay in the conservative side in case the flange configuration and the selected methods lead us to apparent inconsistencies (see discussion in section 3.1).

It shall be of major importance to predict accurately the local buckling load of that laminated plate (with one side free, the other supported in skin), in order to not having excess of conservatism in margins, or over-sizing (over-weight) in the parts. Expressions in sub-chapter 2.1 permit to take into account cases in which the tee web is supported at its ends (e.g. riveted or joined to ribs or frames). More even, if those supports are sufficiently rigid (e.g. tee stringer and ribs shear ties joined as one integral part), ends could be considered clamped. Taking benefit of a skin support harder than the simply-supported condition is more complex, and typically not possible without detailed numerical simulation and/or test validation.

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4 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
a	mm	Length of the column, or of the long side of a section segment (typ. $a \gg b$)
A	mm ²	Area of a segment of the cross section profile
$[A]$	N/mm	In-plane stiffness matrix of a composite laminate
b	mm / -	Width of section segment (flange), or, as subscript, relative to buckling
c	-	As subscript, relative to compression
cc	-	As subscript, relative to crippling
$[B]$	N	Matrix defining the laminate shear-bending coupling
$[D]$	Nmm	Bending stiffness matrix of a composite laminate
E	MPa	Elastic Young modulus of the material or of the laminate
ε	-	Strain
f	-	As subscript, relative to in-plane bending
F / σ	MPa	Stress
F_{xc}	MPa	Compression ultimate apparent stress of a stable laminate (plain compression strength)
I	mm ⁴	Bending moment of inertia
i	-	As subscript, refers to the identification number of a section segment
M_b	Nmm	In-plane bending moment of the segment at buckling onset
N_{xb}	N/mm	Compression buckling load-flow (force per unit width) of the segment
N_{xyb}	N/mm	Shear buckling load-flow (force per unit width) of the segment
OEF / 1EF	-	One Edge Free
NEF	-	No Edge Free (all sides supported)
P	N	Crippling allowable force of the cross-section
R	-	Failure index, i.e. ratio between uni-axial load and allowable load
s	-	As subscript, relative to shear
t	mm	Thickness of section element (flange)
u	-	As subscript, relative to ultimate strength, within this scope, to plain strength in a stable section
UD	-	Unidirectional, refers to materials with the fibre reinforcement just in one direction

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REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[S. G. Lekhnitskii]	1984	Anisotropic Plates
Ref. 2	[SAE] CMH-17	2012	Composite Materials Handbook (Volume 3)
Ref. 3	ESDU80023	1995	BUCKLING OF RECTANGULAR SPECIALLY ORTHOTROPIC PLATES
Ref. 4	[C. Kassapoglou]	1st Ed.	Design and Analysis of Composite Structures
Ref. 5	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 6	[E. E. Spier]	1978	Stability of Graphite/Epoxy Structures with Arbitrary Symmetrical Laminates
Ref. 7	NT-T-ADS-93054	Ed. B	CRIPPLING EN LARGUERILLOS DE MATERIALES COMPUESTOS
Ref. 8	NT-T-ADS-97013	Ed. A	Manual Teórico del Programa ARPA, version 2.0
Ref. 9	[M. P. Nemeth]	1997	Buckling Behavior of Long Symmetrically Laminated Plates Subjected to Shear and Linearly Varying Axial Edge Load
Ref. 10	[P.M. Weaver, J.E. Herencia]	2007	Buckling of a Flexurally Anisotropic Plate with One Edge Free
Ref. 11	[Herencia, Weaver, Friswell]	2010	Closed-Form Solutions for Buckling of Long Anisotropic Plates with Various Conditions under Axial Compression
Ref. 12	HSB45111-10	Is. A	Buckling of symmetrically layered orthotropic plates with free and elastically clamped edges"
Ref. 13	[P.M. Weaver]	2012	On optimization of Long Anisotropic Flat Plates subject to Shear Buckling Loads
Ref. 14	[Tarján, Sapkás, Kollár]	2002	Local Buckling of Composite Beams
Ref. 15	TEA-T-NT-210008	Is. A	CRIPPLING OF COMPOSITE STRINGERS DISCUSSION ON TESTS AND ANALYTICAL METHODS

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 1. Software programs referenced in this chapter

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RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
1.0 04/11/2019	F. Glock G. Baños	First issue	All pages
2.0 09/03/2021	G. Baños	Cover flexural/twist anisotropy and restrained boundaries in compression buckling calculations Crippling method for composite segments updated References and miscellaneous updates	Pages 4-8 Pages 11-12 Pages 2, 10, 15-17

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Methods (TEPAP-TL4)	G. Baños 09/03/2021	Florian Glock 09/03/2021

Table 3. *Approval signatures table for last revision of this chapter*